Expressions

Letters stand for numbers.

Exponents: how many times to multiply

$$3^4 = 3 \times 3 \times 3 \times 3 = 81$$

The small 4 says: use 3 as a factor 4 times. 3^2 is “3 squared” = 9.

Write it with a letter

5 more than n	$n + 5$
7 less than n	$n - 7$
3 times n	$3n$
n split into 4 equal parts	$n \div 4$

Find the value: put the number in for the letter

When $n = 5$: $3n + 4 = 3 \times 5 + 4 = 19$

Same value, different look

$2a + 5a = 7a$

$3(x + 4) = 3x + 12$

Your turn!

When $n = 6$: $2n + 7 =$ **19**

For teachers

Goal: exponents, writing and reading expressions with letters, finding their value and equivalent expressions (6.EE.1–4). Say “3n means 3 times n. Put the number in for the letter.”

Expressions: Practice

A letter stands for a number.

Part 1 Exponents Write it as multiplication, then find it.

1) $2^5 = 2 \times 2 \times 2 \times 2 \times 2 =$ **32**

2) $3^3 = 3 \times 3 \times 3 =$ **27**

3) $5^3 = 5 \times 5 \times 5 =$ **125**

4) $4^3 = 4 \times 4 \times 4 =$ **64**

5) $6^2 = 6 \times 6 =$ **36**

6) $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$ **64**

7) $7^2 = 7 \times 7 =$ **49**

8) $3^4 = 3 \times 3 \times 3 \times 3 =$ **81**

9) $9^2 = 9 \times 9 =$ **81**

10) $8^2 = 8 \times 8 =$ **64**

11) $2^4 = 2 \times 2 \times 2 \times 2 =$ **16**

Expressions: Practice (continued)

Part 1 Exponents (continued)

12) $5^4 = 5 \times 5 \times 5 \times 5 =$ **625**

Part 2 Write it with a letter Fill in the number.

13) 6 more than n: n + **6**

14) 9 less than n: n − **9**

15) n times 8: **8** n

16) n shared by 5: n ÷ **5**

17) 12 more than t: t + **12**

18) 4 less than y: y − **4**

19) double k: **2** k

20) 10 times m: **10** m

21) p split into 3 parts: p ÷ **3**

22) 15 added to r: r + **15**

Expressions: Practice (continued)

Part 3 Find the value Put the number in for the letter.

23) when n = 5: 3n + 4 = **19**

24) when n = 9: 2n + 7 = **25**

25) when n = 6: 5n + 3 = **33**

26) when n = 8: 4n + 1 = **33**

27) when n = 3: 6n + 10 = **28**

28) when n = 4: 7n + 5 = **33**

29) when n = 2: 8n + 9 = **25**

30) when n = 5: 9n + 6 = **51**

31) when n = 12: 2n + 11 = **35**

32) when n = 11: 3n + 8 = **41**

33) when n = 7: 4n + 12 = **40**

34) when n = 9: 5n + 2 = **47**

Expressions: Practice (continued)

Part 3 Find the value (continued)

35) when n = 8: 6n + 7 = 55

36) when n = 4: 10n + 13 = 53

Part 4 Put like terms together No help this time.

37) 2a + 5a = 7 a

38) 4x + 3x = 7 x

39) 6y + 7y = 13 y

40) 3n + 9n = 12 n

41) 8b + 4b = 12 b

42) 5k + 6k = 11 k

43) 7m + 8m = 15 m

44) 9p + 3p = 12 p

45) 10t + 5t = 15 t

46) 11z + 4z = 15 z

Expressions: Practice (continued)

Part 4 Put like terms together (continued)

47) $6c + 12c = \boxed{18}\ c$

48) $13r + 2r = \boxed{15}\ r$

Part 5 Open the parentheses Multiply each part.

49) $3\ (\ x + 4\ ) = 3x + \boxed{12}$

50) $5\ (\ y + 2\ ) = 5y + \boxed{10}$

51) $2\ (\ n + 9\ ) = 2n + \boxed{18}$

52) $6\ (\ a + 3\ ) = 6a + \boxed{18}$

53) $4\ (\ m + 7\ ) = 4m + \boxed{28}$

54) $7\ (\ b + 5\ ) = 7b + \boxed{35}$

55) $8\ (\ t + 2\ ) = 8t + \boxed{16}$

56) $9\ (\ k + 3\ ) = 9k + \boxed{27}$

57) $2\ (\ p + 11\ ) = 2p + \boxed{22}$

58) $10\ (\ r + 6\ ) = 10r + \boxed{60}$

Expressions: Practice (continued)

Part 6 More values

59) when n = 3: 12n + 16 = **52**

60) when n = 2: 9n + 19 = **37**

61) when n = 6: 4n + 8 = **32**

62) when n = 10: 11n + 12 = **122**

63) when n = 3: 11n + 19 = **52**

64) when n = 5: 10n + 13 = **63**

65) when n = 12: 11n + 9 = **141**

66) when n = 8: 7n + 5 = **61**

67) when n = 3: 12n + 17 = **53**

68) when n = 2: 6n + 10 = **22**

69) when n = 4: 8n + 4 = **36**

70) when n = 6: 6n + 1 = **37**

Expressions: Practice (continued)

Part 6 More values (continued)

71) when n = 12: 8n + 3 = 99

72) when n = 6: 4n + 5 = 29

73) when n = 2: 11n + 2 = 24

74) when n = 12: 8n + 2 = 98

75) when n = 4: 12n + 3 = 51

76) when n = 10: 7n + 14 = 84

Part 7 More like terms

77) 9y + 3y = 12 y

78) 11g + 7g = 18 g

79) 8g + 6g = 14 g

80) 12x + 10x = 22 x

81) 8p + 13p = 21 p

82) 9t + 2t = 11 t

Expressions: Practice (continued)

Part 7 More like terms (continued)

83) $4c + 7c = \boxed{11}\ c$

84) $14c + 6c = \boxed{20}\ c$

85) $4w + 4w = \boxed{8}\ w$

86) $14k + 15k = \boxed{29}\ k$

87) $12a + 13a = \boxed{25}\ a$

88) $11y + 5y = \boxed{16}\ y$

Part 8 More parentheses

89) $2\ (\ c + 3\ ) = 2c + \boxed{6}$

90) $4\ (\ y + 15\ ) = 4y + \boxed{60}$

91) $2\ (\ d + 4\ ) = 2d + \boxed{8}$

92) $5\ (\ a + 5\ ) = 5a + \boxed{25}$

93) $10\ (\ q + 13\ ) = 10q + \boxed{130}$

94) $5\ (\ p + 10\ ) = 5p + \boxed{50}$

Expressions: Practice (continued)

Part 8 More parentheses (continued)

95) $6(w + 10) = 6w + \boxed{60}$

96) $4(y + 7) = 4y + \boxed{28}$

97) $11(n + 14) = 11n + \boxed{154}$

98) $11(w + 4) = 11w + \boxed{44}$

Part 9 Remember? Problems from earlier packets.

99) $-16 \boxed{<} -6$

100) $|-37| = \boxed{37}$

101) GCF of 56 and 84 = $\boxed{28}$

102) LCM of 12 and 15 = $\boxed{60}$

103) $5.46 \div 0.6 = \boxed{9.1}$

104) 80% of 35 = $\boxed{28}$

Expressions: Practice (continued)

Part 10 Word problems Write an expression, then find its value.

105) Movie tickets cost $9 each. Write the cost of t tickets. What do 7 tickets cost?

Number sentence 9t, 9 × 7 = 63

Answer $63

Draw or show your work

106) Mia is 4 years older than her brother, who is b years old. How old is Mia if b = 8?

Number sentence b + 4, 8 + 4 = 12

Answer 12 years

Draw or show your work

107) A pizza is cut into 8 slices. How many slices are in p pizzas? Find it for p = 6.

Number sentence 8p, 8 × 6 = 48

Answer 48 slices

Draw or show your work

108) A square has sides of 9 cm. Its area is 9^2 square cm. What is the area?

Number sentence 9 × 9 = 81

Answer 81 square cm

Draw or show your work

109) Challenge: A taxi costs $3 plus $2 a mile: 2m + 3. What does a 12-mile ride cost?

Number sentence 2 × 12 + 3 = 27

Answer $27

Draw or show your work

Check-In: Expressions

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Exponents

1) $3^5 = 3 \times 3 \times 3 \times 3 \times 3 =$ **243**

2) $10^2 = 10 \times 10 =$ **100**

Part 2 Find the value

3) when n = 7: 3n + 2 = **23**

4) when n = 4: 5n + 9 = **29**

5) when n = 8: 2n + 13 = **29**

Part 3 Put like terms together

6) 4w + 11w = **15** w

7) 12e + 3e = **15** e

Part 4 Open the parentheses

8) 4 (x + 9) = 4x + **36**

9) 3 (n + 12) = 3n + **36**